9B Lesson 5

Growing crops

Year 9 Science · 9B Plant growth · Student Book pages 28 to 31

Name _____ Class _____ Date _____

Today I can:

- ☐ I can explain how fertilisers, pesticides, greenhouses and machines increase crop yield, and how pests and population growth change the food supply.
- ☐ I can describe how new varieties are made by cross-breeding and selective breeding, and interpret the results of a fertiliser investigation.

Key words

yield the amount of useful product a farmer gets from a crop
fertiliser a substance added to soil that contains mineral salts to help plants grow
manure animal waste used as a natural fertiliser
decomposer a microorganism that breaks down dead material and waste
pesticide a substance that kills pests (organisms that damage crops)
insecticide a pesticide that kills insects
fungicide a pesticide that kills fungi that cause plant diseases
herbicide a pesticide that kills weeds
weed a plant growing where it is not wanted, competing with the crop
variety a group of plants bred to have certain characteristics
selective breeding breeding only from the plants with the wanted characteristics, generation after generation

● Core Everyone does this section.

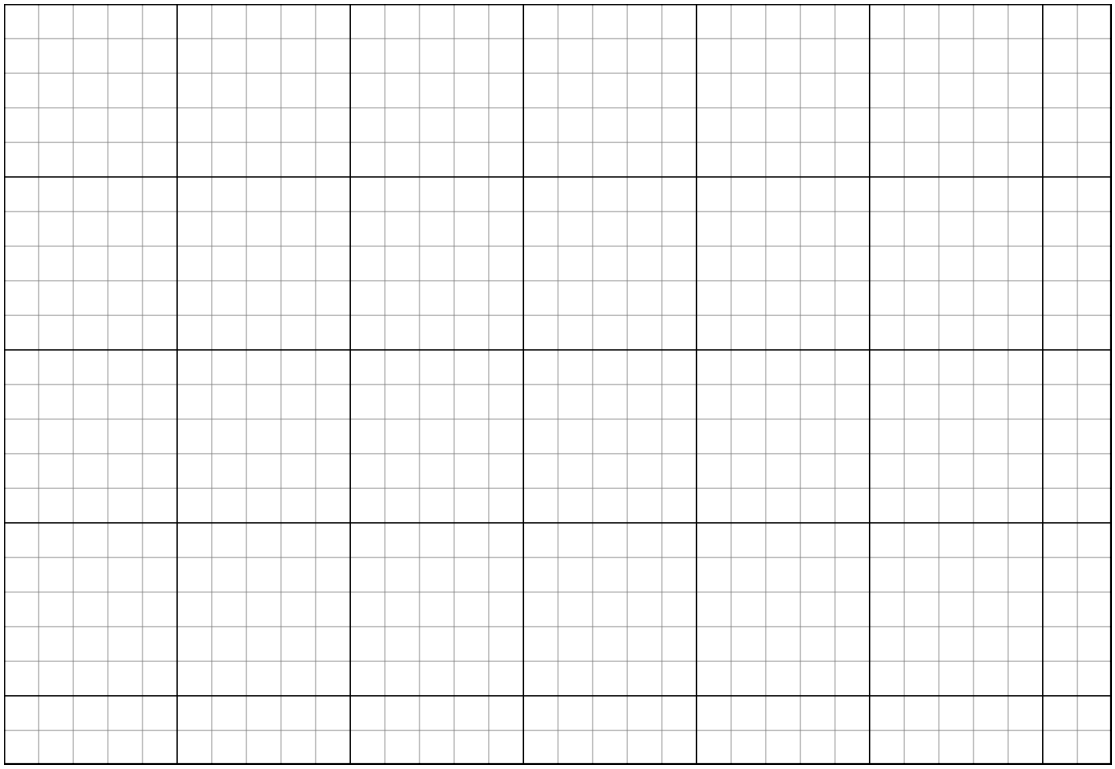
1. Match each type of pesticide to what it kills. Draw a line.

insecticide	weeds with broad leaves, but not narrow-leaved crops such as wheat
fungicide	insect pests such as aphids
herbicide	all weeds (plants competing with the crop)
selective herbicide	fungi that cause plant diseases such as rust

2. Students investigated fertiliser. Each group grew 20 bean plants in identical pots, one plant per pot, and added a different number of drops of the same liquid fertiliser. After a week they calculated the mean increase in height.

a) Plot the results as a bar chart on the grid. Label both axes.

Group	Drops of liquid fertiliser	Mean increase in height (cm)
1	2	4.6
2	4	6.3
3	6	7.8
4	8	7.2



b) Describe the trend in the results between groups 1, 2 and 3.

c) What advice would you give a farmer about how much fertiliser to use, based on the result for group 4? Give a reason.

3. Complete the sentences. Use the word bank.

Fertilisers contain _____ _____ that help plants grow. Farmers use cheap _____ fertilisers and natural ones such as _____ (animal waste).

Microorganisms called _____ break down manure and release the mineral salts, but this takes a long time.

●● Stretch When Core is finished.

S1. Aphids are insect pests. They suck the sugary liquid (sap) from the phloem of plants. Explain how spraying a crop to kill aphids could increase the crop yield.

S2. Selective herbicides kill weeds with broad leaves but not crops with narrow leaves.

a) Suggest why a wheat farmer might want to use a selective herbicide.

b) Suggest why farmers in low-income countries may not use selective herbicides, and what they might do instead.

S3. Describe one change you would make to the method of the fertiliser investigation to be more sure of the results. Give a reason for your answer.

●●● Challenge Explain in full sentences.

C1. An apple breeder has a yellow apple variety that is not sweet and a red apple variety that is sweet. Describe how the breeder could produce a new yellow, sweet variety. Use the words cross-breeding, select and generations.

C2. Farms in the UK today produce 35% more food per square metre than they did in 1973. Explain three different ways farmers have achieved this.

My thinking (any language)

Before a farm is allowed to expand into a wild area, a plant ecologist carries out an ecological audit. Suggest three pieces of information the ecologist should record, and explain why each one matters.

Word bank mineral salts · artificial · manure · decomposers · yield · fertiliser · pesticide · insecticide · fungicide · herbicide · weed · variety · selective breeding

✂ ---------- cut here

Exit ticket 9B Lesson 5: Growing crops Name _____

1. What do fertilisers contain that helps plants grow?

2. What do herbicides kill, and why does this help the crop?

3. Describe one step in selective breeding.

Today I can: Goal 1 ☐ Goal 2 ☐